Clinevo Smart Inbox Assistant

Technical Submission Write-Up

AI-Assisted Healthcare Email/PDF Intake, Classification & Human Review

CANDIDATE	DELIVERABLE SCOPE	TECH STACK
Sri Hari	Core Intake (100%) + Literature Screening Bonus	Angular 18 • Spring Boot 3 • Python / FastAPI

1. What I Built

The Clinevo Smart Inbox Assistant is a prototype that receives synthetic healthcare emails and PDF attachments, classifies each communication, extracts relevant facts, links those facts back to verifiable source passages, and presents the result to a human reviewer for confirmation or correction.

Safety Report (ICSR)	Quality Complaint (PQC)
Adverse event experienced by a patient.	Physical or packaging defect with a product.

Medical Information (MI)	Not Relevant
Guidance inquiry with no adverse event or defect.	Newsletters, marketing, or unrelated correspondence.

Multi-label cases are explicitly supported — for example, a contaminated vial that also caused an adverse reaction is classified as both PQC and ICSR.

2. Architecture

Email / PDF
↓
Spring Boot Ingestion
↓
Python AI Service
- Document Understanding
- Classification
- Fact Extraction
- Evidence Verification

↓
Spring Boot API / Persistence
↓
Angular Reviewer UI
↓
Human Reviewer

3. Processing Flow

Receive → Understand → Classify → Extract
↓
Ground with Evidence → Verify → Human Review → **Audit**

4. Key Engineering Decisions

Decision	Why	Result
Bounded document processing instead of global vector RAG	Intake packages are small; chunking would separate patient details from suspect drugs.	Entire document is passed in-prompt with layout preserved; avoids document fragmentation caused by arbitrary chunk boundaries.
Category-specific payloads	Forcing all messages into an ICSR schema produced empty tables and false warnings.	Distinct payloads per category give a clean, relevant reviewer view and reduce schema confusion.
Evidence-linked fact model	Reviewers need to see where an extracted value came from before trusting it.	Extracted facts include source citations where available, providing location and verbatim snippet.
Separate retrieval from semantic verification	High lexical similarity does not guarantee a passage actually supports a fact.	Retrieval and NLI-based verification are split into two steps; reduced false evidence linkages observed in earlier iterations.
"Not stated" for missing information	General-purpose models tend to infer unstated clinical details.	Prompts enforce negative constraints; missing values are marked "Not stated" instead of guessing.
Fixture + live mailbox ingestion	Reviewers and tests need to run offline without live email credentials.	An ingestion abstraction supports both live IMAP polling and offline fixtures.
Human-in-the-loop review	Autonomous decisions are inappropriate for patient-safety-relevant content.	Human review remains required because model outputs can contain errors or omissions. The AI prepares a draft case; the human reviewer confirms or overrides it.

5. AI / Prompting Approach

- **Structured outputs:** Enforced via structured schemas to reduce output parsing issues.
- **Category-aware extraction:** Tailored to the detected communication type (ICSR, PQC, or MI).
- **Source-grounded facts:** Extracted facts include source citations where available.
- **"Not stated" for missing info:** Negative constraints avoid guessing when data is absent.
- **Multilingual handling:** Handles non-English communications (e.g. German, Spanish).
- **Tables / scanned documents:** Layout-aware parsing and vision models handle complex documents.
- **Separate evidence verification:** Two-stage verification checks passage support (SUPPORTS / CONTRADICTS).
- **Literature screening bonus:** Screens journal PDFs, rejecting negative controls and splitting multi-patient cases.

Safety note: Model outputs can contain errors or omissions. The prototype is designed as a reviewer aid; the final decision remains with the human reviewer.

6. Reviewer Workflow

SOURCE
Email / PDF / Image

→

STRUCTURED REVIEW BRIEF
Category • Summary • Facts • Evidence

→

REVIEWER
Confirm • Edit / Override • Flag

Confirm | Edit / Override | Flag

The reviewer workspace pairs the source document with a structured review brief so extracted facts can be verified in context. Clicking an evidence reference navigates to the relevant source region and provides document-level evidence highlighting where supported. The reviewer can confirm the draft case, edit or override any field, or flag it for further attention. Reviewer actions are recorded in the timestamped audit history.

7. Testing & Results

Evaluation Area	Measured Prototype Result
Primary triage classification	**27 / 27 (100%)**
Multi-label detection	**100% on benchmark case**

Evaluation Area (Continued)	Measured Prototype Result
Core fact extraction	**94.8%**
“Not stated” handling	**No benchmark hallucinations observed**
Defect photo flagging	**100%**
Literature negative controls	**100%**
Literature case splitting	**3 / 3**
AI service tests	**114 / 114 passed**
Angular tests	**56 / 56 passed**
Spring Boot tests	**20 / 20 passed** (7 core + 13 integration)
Live mailbox ingestion	**Verified end-to-end**

Evaluated on a synthetic benchmark of 27 cases spanning emails, PDFs, and images. Results reflect prototype-stage evaluation, not a claim of production-grade certainty.

8. Important Engineering Fixes

- **IMAP Fetching Optimization:** Ingestion was optimized to index baseline UIDs and timestamps at startup, ignoring past mailbox clutter and ingesting only live emails.
- **Canonical Case Identifiers:** Stable canonical case identifiers (CASE-01 through CASE-12) were decoupled from database auto-increment keys.
- **Retrieval vs. Semantic Verification Separation:** Lexical retrieval and semantic NLI verification were separated into distinct stages; reduced false evidence linkages observed in earlier iterations.
- **Missing-Value Evidence Isolation:** Fields marked “Not stated” were isolated so they do not receive unrelated document text passages as evidence.
- **Category-Specific Schemas:** Distinct payloads were designed per category, reducing output parsing issues through structured schemas and removing irrelevant fields.
- **Verification Batching:** Fact verification requests were batched; reduced unnecessary API calls and rate-limit pressure.
- **PDF Viewer / Highlighting Improvement:** Coordinate-based document navigation was enhanced to direct reviewers to relevant source regions where supported.

9. Limitations

Prototype Boundary	Future Production Work
Synthetic data only	Controlled real-data / privacy architecture
Cloud model dependency	Multi-provider fallback strategy
Local H2 evaluation database	Managed enterprise database
No automatic MedDRA / WHO Drug coding	Dictionary integration
Prototype audit history	Formal validation and operational controls
PDF viewer limitations	Document rendering / annotation layer improvements
Human review required	Workflow / identity controls for production

10. Production Next Steps

- Data privacy and de-identification
- Enterprise authentication and access control
- Production database and distributed processing
- Formal validation and operational controls

11. Final Takeaway

> *“This prototype demonstrates an end-to-end AI-assisted intake workflow for healthcare communications. It reduces first-pass manual work by organizing incoming messages, extracting relevant information, and linking facts back to source material. The reviewer remains responsible for confirmation and correction. The prototype was evaluated using synthetic benchmark data and is not presented as a production regulatory system.”*

GitHub: https://github.com/sriharizz/smart-inbox **Run locally: run.bat / ./run.sh**